[00:00] 发言人 1: So here, this is not odds is the ratio. So this is not odds. As a fine function, so what’s the difference between a fine function and a linear function? So a fine function is linear function plus a bias term. Which is a linear decision boundary, and here we can let’s say trees.

[00:31] 发言人 1: This as they will have here why is equal plus one given x equal to one over one plus e to the negative z right and similarly why is equal to negative one given x this to or big this should sum up to one then we have e to the negative z plus one. Over one plus a negative C and we can compute the relation. negative one given x is equal to one over into the negative z which is equal to e to the z and we take off those. This gives you. Z 为其 Z 或者 W transport X plus B 跟其 Z 是一回事。

[01:39] 发言人 1: Then the classification function is among P O Y is equal to K K can be. Can take values from the one one events. The model prime actors generally have no close from solution, and can only be estimated numerical optimization.

[02:04] 发言人 1: So opposition much of machinery can be written as an opposition problem. We minimize a loss function computed on this is d, which is parameterized by theta. Theta here is a prime prime prime trajectory of the organization.

[02:23] 发言人 1: And we can decompose this loss function into the sum of the loss function divide on individual samples. Here is the loss on the ice train example, which is by the same sit down and some overboard and train examples. And R is a regularizer to prevent overfitting the data set B.

[02:48] 发言人 1: So in not just regression, we have C now is equal to. The weight factor plus the variance b and the individual loss function is indeed the. Modest loss, right? And RC 它 is one over C W transports W.

[03:18] 发言人 1: And types of donation，but convex of donation，which the easy case include not just regression in your regression as well，well non convex of donation, it’s hard in general includes deeper，and back to let’s say one hundred or two hundred years ago，we don’t categorize operation this way，we categorize。operation so linear operation. versus nonlinear relation.

[03:55] 发言人 1: population. The researchers. Opposition began to find that a portion of long linear optimization is easy to solve to its global optimal, and they find their common characteristics, and then retire right operation into convex operation, non convex operation, and linear optimization is for sure convex.

[04:31] 发言人 1: So let’s introduce the first fundamental concept in complex opposition, that is nice. Nine segment between two point x and y consists all points three which is here. This is known as a convex computation.

[04:56] 发言人 1: Because we constrain alpha to be zero and one, if we don’t constrain alpha. We close this an inner competition. If we constrain our to be zero and one, we call it convex combination.

[05:15] 发言人 1: And convex set, a set is convex if it contains entire line segment join any two points, any two of its points, meaning that here this are. Randomly chosen from taxi for me to randomly chosen coins where alpha is between zero and one we

have this is also in the set this is one of the topics. So how to prove it is to be conquered.

[06:02] 发言人 1: But you just find one single counter example or particular error points. And some portion of the nine segment are what be on the set that is done. And here’s an example. We have one convex to non convex. And I said the first is convex because for any two points the nine segment.

[06:31] 发言人 1: Is intersect，and for the same example，of course you can find two points。Where is the nice segment just set? But to the part which must let this culture be valid for all pairs of points, right? And in particular for this pair of points, you see that a portion of the nice segment does not belong to the set.

[06:59] 发言人 1: And then we can disapprove it. It’s not convex. And for the third particular example.

[07:09] 发言人 1: 嗯，the boundary is open not close，right? So if we take this two exam，take this two points and draw the nine segments，this part by out of the set, so it’s also not complex。This is just for my definition.

[07:30] 发言人 1: Hyperplanes and half spaces. So hyperplanes says of the form x. Satisfy a transport x plus b where a is not equal to zero is a normal vector, so.

[07:47] 发言人 1: Right angle here and half space set of the form x such that a transpose x is less than or equal less than or equal to b where. zero. And it contains if we, if we define a to point along this direction, so this house base is is shaded area.

[08:16] 发言人 1: So why is that? 所以，We can draw point here，right? We can draw point let’s say X。啊，不对。With this, and we simply compute a transpose x minus x not.

[08:38] 发言人 1: And this is equal to a transpose x minus a transpose x not what’s a transpose x not since x not is x zero is on the plane right we have a transpose x zero is equal to b. right it’s on the plane. Therefore, this is equal to a transpose x minus b, and we see that we have an angle larger than.

[09:08] 发言人 1: At the degree and this in your product should be. This than zero and they transport it’s this than b that’s in it’s in the set and if you find a point on this side so the inner product is going to be positive it’s not in the set. 嗯，so hyperplane and house space are convex。And now let’s prove prove it.

[09:41] 发言人 1: So let’s first draw two points, right? X and Y, which is in the set with we prove half space are compact, which is in set, meaning that those two points satisfy this inequality. And we work with alpha with zero one, and we examine convex combination. 阿尔法 X plus one minus 阿尔法 Y.

[10:16] 发言人 1: one minus r by one and we. Test whether this point is also in the set or not. That’s why we add a transpose in front, and this gives alpha, a transpose x plus one minus alpha, a transpose y, and this is this alpha b plus one minus alpha b.

[10:39] 发言人 1: This is equal to b, then we also prove that. This is this sound or equal to B meaning that this is also in the set. 所以你开始去出发。给。

[11:03] 发言人 1: The operations are preserved complexity. So practical methods for establishing complexity of a set. The first is to apply definition where we randomly choose two points and the world is in our part that which is from zero one.

[11:23] 发言人 1: We prove this also in the set. And we can show that axes often from simple convexes at half space have plans by operations that preserve complexity such as intersection and a fan mapping and so on. Now convex function，so if the function from R^n，it takes the n-dimensional vector as input，and gives you a number，the output is just a number，R，is called convex if well。

[12:00] 发言人 1: Two conditions: if its input domain is complex set, and. It satisfies the equality f alpha x plus one minus alpha y is not equal to alpha f x plus one minus alpha y for all x y is in domain alpha and zero and one, and pictorially to understand this equation. well，it’s the experience。那这些时间是还有。we have here y and this gives you f one and this we work with.

[12:47] 发言人 1: 对，come with combination，this is alpha x plus one minus alpha y, and we have this。as if of 阿尔法 X plus one x。啊，法官。then let’s connect.

[13:17] 发言人 1: If x and if y and further extra is mine. 那谁是？Point 就是刚才这一种。

[13:30] 发言人 1: 阿尔法，FX plus one minus 阿尔法，and this equally test you out. We always gonna have this app here. The gap is greater than zero。AND THE OTHER WORDS.

[13:51] 发言人 1: 是在那一 segment. Join F X X F X Y F Y is always gonna lie above. The curve, right? And for non convergence, we typically have three types of points that share the same first recognition as first local minimum. Local minimum is the minimal around its neighborhood. We have global minimal, global minimal is the minimal over the whole input domain, and have saddle points which is neither minimal nor maximum, but share the same first order characteristics that that is the gradient respect to global minimal, local minimal and saddle points are zero.

[14:52] 发言人 1: And if it’s concave, you negative f is convex. And for concave function, we just for concave function, we just reverse the inequality. So this gives you function.

[15:12] 发言人 1: we have several examples on one dimensional function like a function exponential function power function on r plus plus so this means. strictly. Positive, what’s R plus R plus minus now? Negative are this real number and our powers of absolute value have negative entropy.

[15:52] 发言人 1: I also have compute function that a five powers. No grace, no grace function. So as you see in this time. We have a particular function for fun function, right? That is both.

[16:12] 发言人 1: 好，不要。嗯，compute，然后 do you can you identify another function that is those compute and complex。Another type of function, what the fine, that’s a close come, that’s not come here.

[16:36] 发言人 1: And how do you think about this question conceptually? So for convex function, we need more directions to curve up a little bit. So that it has a

global minimum right for continuous function with its all directions to close down a little bit so that it has a global maximum.

[17:00] 发言人 1: And we cannot have a curve that slows curve up and curve down, and only compromise we can make. Is that the curve has no curvature? It’s a straight line.

[17:15] 发言人 1: It’s a straight plane. Let’s say a function. So the function is the only type of function. I suppose convex and concave. we have examples on our end.

[17:33] 发言人 1: So profile function a transpose x plus b we have long where l infinity which is not just the maximum of that absolute value we can define functions on matrices we have profile function. which is equal to the trace of the transpose as b if this is the trace. Twice of a square matrix is simply the sum.

[18:08] 发言人 1: of its elements. And I encourage you to identify the equivalents of this two double summation with this lift matrix of you. Right, you can practice my patient is.

[18:29] 发言人 1: A LOT OF A LOT OF PRODUCTS. Right, you can, you can think of this diagonal term of this result. FC is the inner product between the first row obey and first row.

[18:46] 发言人 1: Of x and the second angle in the first row, second row of b and the second column, and second row of of x because we take the transpose of a and they. With some more dynamic terms. The second is spectral form or maximum singular value.

[19:09] 发言人 1: In this case, this is not not realized the outlaw because outlaw only works on vector if the input is a matrix. They call it spectral, which is equal to the maximum singular value, which is equal to. The square root of the maximum eigenvalue, this is the singular value.

[19:33] 发言人 1: 哦。类似的 I 跟 M。I can buy the of it transmission and if you don’t know don’t know the definition of singular value like that’s fine we will get to cover the when we talk about component analysis or PCs for short but for now.

[19:59] 发言人 1: 有。You can understand this simply as this is a very complex function because to compute the maximum eigenvalue we may involve some numerical calculation interrupting methods. But still, the function is complex, you should be able to solve the equation.

[20:22] 发言人 1: And now let’s prove the convex of some function. This is a simple correct function. And we know that the domain input domain domain of f is r, r is convex set.

[20:39] 发言人 1: And secondly, prove this in quality. So we have alpha x plus one minus alpha y squares is equal to alpha. We expand the squares.

[20:50] 发言人 1: is equal to alpha squares x squares plus two alpha one minus alpha x one plus one minus alpha squares by squares, and what we really need. Is alpha squares and one minus alpha y squares the converse calculation right over the x and

one and that’s why we add. that’s why we add f x alpha x here and matters in alpha alpha x here and we expand this into.

[21:25] 发言人 1: ONE MINUS 阿尔法 MINUS 阿尔法 ONE MINUS 阿尔法。啊人。We can combine. These two terms, we can combine these two terms. To get negative alpha one minus alpha squares we have also negative alpha one minus alpha we have alpha one minus alpha we take the common terms out we have negative alpha one minus alpha.

[22:00] 发言人 1: X squares plus y squares minus two xy and the the first two term is just alpha x squares one minus alpha y squares and we know that. nice. This is equal to x minus y squares which is greater than zero one max alpha is greater than zero alpha is greater than zero if we take they connect to sign the whole term. Is this and zero and this justify this inequality and we are done.

[22:52] 发言人 1: And to prove the complexity of absolute function is easy, right? Equal domain R is power set, and we can just simply using this triangle inequality. And since alpha and one plus alpha is non negative, we can take the two values of the absolute function.

[23:18] 发言人 1: exponential function。The proof of convexity we hear without loss of generality we also makes strictly greater than one. The input domain is a compact set. And then.

[23:34] 发言人 1: The equality e to the alpha x my plus one minus alpha y is equal to e to the y times e to the alpha x minus one. And we do Taylor expansion over this term, which gives you the sum of n from zero to positive infinity. one over factorial r up to the n x minus y to the n and we can.

[24:02] 发言人 1: Example，the first term，n is equal zero，this is equal one，right? one minus zero factorial up to the n，x minus one for n is equal to one，well，one，so we only exam this term。Well one plus the sun and 以及抠图 one or two infant.

[24:27] 发言人 1: 很不错啊，阿尔法三。And since alpha is in zero and one, we have alpha is greater than or equal to alpha squares is greater than or equal to alpha q alpha n, right? And this this term this term is this angle to one plus the sum and from one to put infinity or the natural very place.

[25:00] 发言人 1: S 阿尔法 N with 阿尔法 according to this property。And x minus y and we take the alpha out of the summation in this term。这个高速吧。阿尔法 N plus N 几个图案，或者平均。

[25:32] 发言人 1: 啊，能。See races, the study index is one right? We want to we want to work with a study index is zero. We replace one with zero which corresponds to add add one alpha and we need to minus one alpha to compensate for it. And this is one better server, and this we do the. There is expansion in the reverse order, which gives you alpha e to the x minus one, and this is equal to one minus alpha into y plus alpha into x, and well done.

[26:21] 发言人 1: The properties of a convex function, any line segment join to points on the ground lies on or above the curve. Right? For this one, we know that this nice above the curve.

[26:36] 发言人 1: What if the function by itself is a fine? 对呀，So the two points unless others. Alright，okay，the second average of minimum is a local minimum，that’s why comics，okay，as long as you find a minimum，must be. And we will prove this one next lecture. Uh, and this is why this is what makes conversation music is surprised by the local minimum as well as those active global.

[27:19] 发言人 1: And properties of convex functions, nonlinear combination of convex functions convex where they be are greater than zero composition of five bag of convex points convex and it is include f a x plus b and pointwise maximum of convex functions is convex.

[27:55] 发言人 1: We have this. 啊，是的，是。This as this is define this as FX this as GX you want this FX GX.

[28:09] 发言人 1: what’s point is maximal for this x. 九一八。We have GX here, since GX is greater than FX, right? We keep this part.

[28:26] 发言人 1: And eventually we will keep this curve. 啊，And this orange she curve is time x as old by the past. And the composition of convex functions most generally convex neural network like this.

[28:50] 发言人 1: And here is a simple example, simple example to verify this. We have FX as convex. Right, we add the matter sign from there to.

[29:05] 发言人 1: You can consider AX or AZ equal to one. AX is compact, is provided, provided is also compact and compact, right? So that way is compact, it’s not compact. Generally, the completion function is later compact, more compact. And we have first order.

[29:27] 发言人 1: Condition f is differentiable if the gradient partial f partial x one partial f partial x two partial f partial x n exists at every point in the domain. And we have first several condition differentiable f is complex domain is complex if and only if. F X is greater than equal to F Y is greater than F X plus gradient F X transports Y.

[30:01] 发言人 1: I understand. And here, what’s the right hand side? Right hand side is essentially.

[30:09] 发言人 1: 嗯，first order。Taylor expansion. 哦。At the point x and x.

[30:35] 发言人 1: Here’s a curve. This curve is f y, and we approximate this f y using first order Taylor expansion at this particular point x f x using f x plus gradient transpose one minus x. 啊，对，一个 global 的 higher order terms。From this curve.

[31:00] 发言人 1: You see that the first order approximation of f is a global underestimate. It’s always gonna be below the curve. It’s not like.

[31:14] 发言人 1: This is not like not like the two nine two the nine segments that join this curve. This curve is going to be above the curve. And this first order of proposition is always going to be in.

[31:38] 发言人 1: First second order condition，the f is twice differentiable if the hessian matrix and this is symmetric。which is. Secondary derivative respect x i x j exist at each x.

[32:01] 发言人 1: And twice differentiable by f is convex domain is convex if and only the hessian matrix is. This read that positive semi definite is a generalization of a positive number for matches. So if a matrix is order to send definite then.

[32:27] 发言人 1: This is for all for all possible x we have a transpose a x transpose a a x is going to zero so this is a number. We can compare number in zero to see whether it’s possible. And this is why this is a generalization of a positive number.

[32:54] 发言人 1: So if we reduce a to one by one. Right, this for all equal x, this is a number we have. A X A X should greater than zero right this is A X squares should greater than zero meaning that A should greater than zero it’s a generalization of the positive number.

[33:26] 发言人 1: So this is the only way to prove whether the matrix called the same different. So positive same different does not necessarily mean that all entries in the matrix are positive. We can find country example that for matrices all entries are positive positive, but it is not a set positive same different matrices.

[33:51] 发言人 1: 咳，喂，你可以 second order condition to the complexity of a provided function，would just take the. Double winnet，which is equal to two，and investment in this。Case the haitian matrix is just a number two we compared against zero convex.

[34:14] 发言人 1: An exponential right? We take the double gradient, which is itself the x, which is greater than zero. Let’s come back.

[34:25] 发言人 1: We may as well prove the logistic loss is convex. Remember, this is a different formula for equation for the logistic loss. F x is equal to log. One plus is the negative x. We first take the. First order derivative this is equal to no of x is one over x right this is one over one plus that two bags and we due to change of gradients. 我一。

[35:00] 发言人 1: Further take the gradient of of the domain of the inside the they know and this gives you negative e to the negative e to the negative x one plus e to the negative x and this is equal to negative one over plus events we multiply. Both you know later and later in place，and then from this we’ve take the second order derivative，and this is equal to negative one over equal x squares of one plus x prime，we need to take the gradient。And one matters, one matters, one cancels we have.

[35:55] 发言人 1: one plus equals one plus equals. As denominator we have e to the x as numerator and we divide by e to the x on both denominator and numerator this gives you one plus equals x one plus equals negative x as shown here and we are done. So why not just regression? It’s easier to solve because it is a complex of the ultimate complex function.

[36:35] 发言人 1: 这一块去搜刮啊。Nice segment complex complex function first order condition，complex function。All right.

[36:50] 发言人 1: 我有点第三。The optimization in machine learning，the goal is to optimize the loss function。Define a merchant problem or private estimation convex optimization goes to minimize convex.

[37:11] 发言人 1: Construction respects the parameters example, let’s say like three to six words. When descent, gd using the following update formula, we update theta by moving towards the negative reading direction with a step size alpha. Alpha is also known as learning rate.

[37:34] 发言人 1: And the Laplacian this with the Laplacian Laplacian is a gradient of the Laplacian. And here is an example, right? Well, Laplacian is a group.

[37:47] 发言人 1: The squares agreement is too thin. We use the alpha is zero one two five and we move. Another crowd this on this place on this place on this place. At recent. Listen first chooses an initial point set zero and repeats and theta at t plus one iteration is equal to theta at t situation minus alpha we have the gradient of l. Evaluate at Sigma at this position.

[38:33] 发言人 1: Which is equal to L sigma t minus one over m sum from i equals one over m the gradient of the gradient computer on the ice example at sigma plus sum and from. this water. You will see that the greatness not scalable with your special plan right so if and is very large.

[39:07] 发言人 1: The hot brilliant is world. We need to compute the brilliant of all individual examples, and then sum up their brilliance. After that, we are able to finally do an update.

[39:29] 发言人 1: Right，and how to stop。They are several。Several ways to stop population.

[39:42] 发言人 1: First is the difference. This is a difference between the consecutive proper updates. The know of the difference of the classic updates is this. Or into some small number or the length the green length is this out into some small number where the angle theta is equal to square root of the sum of the theta j squares then we have. 你要输出二手机，它以及扣除。

[40:33] 发言人 1: 首都是二十四。And the PC epsilon epsilon is a small positive constant ten to the left six. And here’s my question, so can you identify some other stopping criteria? Probably used.

[40:57] 发言人 1: Right，it typically use this to set a maximum iteration number, right? So we can set the maximum iteration number to let’s say two thousand. So after reach two thousand iteration we just stop.

[41:22] 发言人 1: 对。And great distance a negative gradient can indeed decrease the loss function. Here is the simple proof. Let theta be any initial point using the first order expansion for the next point which according to the grading distance is computed by a theta zero theta zero minus r by the grading to the l divided at the theta zero and then we have. 没有四个弯。

[42:00] 发言人 1: is approximately equal to zero plus the first order gradient of zero times zero one times zero and this. It’s essentially the same as first order condition appearing previous sites where we have y is equals to one, x is equals to zero, as showing the first order condition here. It’s the same first order expansion.

[42:40] 发言人 1: And the simply we have the dry high sides, the first order. take it. Expansion of L 西塔 one。At point theta zero l theta zero，and we ignore the higher

order terms。And what theta one minus theta zero according to gradient step is negative alpha，the derivative of l divided theta zero。And we have gradient of l theta zero transpose gradient of theta zero.

[43:24] 发言人 1: This gives you l squared of gradient of theta zero. And hence, if the gradient is not zero and alpha is sufficiently small, why do we need alpha? Why alpha needs to be sufficiently small in order to make.

[43:45] 发言人 1: This approximation as accurate also otherwise we cannot ignore the higher order terms. then we have there’s one minus s zero is equal to alpha squares of the gradient of the zero which is. Strictly negative therefore for sufficient small r 方 l sigma one is indeed an improvement over l sigma zero.say question this book.

[44:28] 发言人 1: And initial point matters. So here we contains a live demo and it does not work on my iPad. And you should, you should be working on your computer screen.

[44:40] 发言人 1: You just click this. And for this final point, we saw. proper learning ways you finally land on land on here if we start here you’ll that land on here if we.

[45:05] 发言人 1: If we start here, we can’t find here. If we start here, we can’t find here. If we start here, we can’t find even over minimum.

[45:16] 发言人 1: So the final convergence point depends a lot on the initial point and to improve position run GD multiple times with different initializations. And keep the solution with no slots for all rounds. Second，the learning rate matters。So if alpha is two B，the GD moves more quickly with a high risk that alpha will never converge。

[45:40] 发言人 1: If alpha is small，GD moves slowly，so slowly that you might just lose patience and never reach minimum。Here we consider this particular function at this demo，if I very quickly shows。好。Small alpha is to move towards the minimum and here shows if we set a large alpha then finally covers there this this and outside.

[46:11] 发言人 1: And finally leverage. So unfortunately, no magical things exist to find the perfect learning rate. You have to test general survival to the best.

[46:25] 发言人 1: And in the area of deep learning, we do have several fancy learning rate scheduler. So I could work at the initial phase of the. population we have a warm what’s known as warm warm up stage as the middle training we have some.

[46:43] 发言人 1: Learning with decay at the last phase of the training we may have this stamping area not to reduce this experiment by zero and so on and you earn several language skippers when we touch on differences. Alright, so let’s take a ten-minute break. We will resume at ten thirty.

[48:07] 发言人 1: 然后。

[48:17] 发言人 1: 你。

[48:23] 发言人 1: 也可以。

[48:32] 发言人 1: 可。我。

[51:15] 发言人 1: 我。哦好。

[51:40] 发言人 1: 什么时候？我签同意然后。

[51:56] 发言人 1: 往下。

[52:14] 发言人 1: 其实就在。

[52:21] 发言人 1: 我们是。

[52:44] 发言人 1: 那个。

[52:56] 发言人 1: 他。

[53:46] 发言人 1: 呃，是的，还是。

[54:01] 发言人 1: 嗯。啊，你说。

[54:20] 发言人 1: 然后这个上门期间检查到一个。七七七七。这是。

[55:22] 发言人 1: 想法可以。

[55:34] 发言人 1: 就从去年就是那信号还有四千四千个训练。然后这个预测说信息模型是。

[55:57] 发言人 1: 对。你其实身体政治性的简单，就现在手写那个。所以的话，自己去调还是快还是快。啊。

[56:44] 发言人 1: 去。

[56:51] 发言人 1: 够宜昌是没有很多企业。就。

[57:27] 发言人 1: Okay，let’s continue。Stochastic with addition to computational costs for GD at each iteration is on the order of M, which grows linearly with M. As they number twenty five, it is prohibited to scale up to huge exit where it can be millions or even billions.

[57:51] 发言人 1: Stochastic with the recent SCD for short reduces competition course at each iteration to essentially constant or one. So at the t situation, we randomly sample single just one for an example, and compute the gradient to update the example here. Sigma t situation is equal to sigma t minus our gradient computed on sigma i x i y i safety.

[58:26] 发言人 1: So one gradient one update one gradient one update is very patient. So in expectation SGD is an unbiased estimate of GD, the expectation of the gradient. 哦，balance set is average of the gradient of orange example is a great gradient of the whole data set，which is used in great distance。Here shows the picture of great distance and stochastic great distance the trajectory.

[59:00] 发言人 1: As a secret distance, the treasure is rather stable, and the stochastic great distance is very noisy. And manage to land on the global meeting. So the advantages of city is easier to fit in memory and competition cost due to a single example in persist and expiration.

[59:27] 发言人 1: And on the bright side, the steps has oscillations that during the peer side, which may help the others get out of bad lock me. If you are dealing with a highly enough complex problem. The disadvantage the other side of the coin the steps are noisy the object does not always decrease as for for each step and it may take longer to achieve convergence.

[59:59] 发言人 1: And it loses the advantage of vector operation as it deal with only a single example at a time. So mini batch gradient descent is a compromise. Unlike SGD, mini batch gradient descent randomly sample a small subset.

[60:25] 发言人 1: 然后，数据和教师情况和计算机阅读到我们的团队，那么这里更新公式变得是说，老师，我创作就是说，团队，而且我发现了伟大的，我计算了迷你班，而且安全，非常，非常，就是 average of the gradient。IN THE MINI BATCH. And mid batch with the distance and experiments are extremely popular.

[61:00] 发言人 1: An extensive use in the early. And when we introduce the early, we will see a lot of population techniques. So we advise with this.

[61:18] 发言人 1: Now, let’s do a summary. Come and set. Any two points in the set with alpha zero one, its compass combination should also be accepted.

[61:30] 发言人 1: Compass function two quite over firstly, the only is compass set and secondly satisfy this important also known as the distance equality. And ways to establish convexity, first order condition, and why is always gonna be greater than equal to FX plus gradient of FX transpose. One better things.

[62:05] 发言人 1: Alright, and the second order。Condition the patient matrix be positive semi definite, and numerical methods, the GD decrease and noise should for learning rate, not scalable to larger datasets, as GD should for faster computes, but with noisy greetings。Made a batch really compromise between GT and STD.

[62:35] 发言人 1: And it is the default method used in different because depending deal with future model data. And you just cannot compute the results. over over 这一。Now let’s do, let’s introduce the power of machines part one. The overview to date we’ve seen one example of discriminative linear classifier.

[63:07] 发言人 1: I adjust preparation which uses a maximum remark learning separating upper plane and today going to this example so for big machines or as a bench for short which is pure geometric approach so for now assume the data is interpretable. meaning that the two classes in the training can be separate by nine like this or hyperplane this is already seen on the house errors did say. And here comes the question, what which is the best separation line? There are many possible solutions, right? 你看出 infinitely many lines to separate this particular data set, so which one is the best? what’s your selection criteria?

[64:09] 发言人 1: So we use the maximum margin principle. We define the distance between the separating line, the black line. And a close point as a margin.

[64:25] 发言人 1: So this is margin point, and this is a margin. The distance is a margin, and dashed is a margin line. More simply, know as a margin.

[64:40] 发言人 1: You can think of this space。As among the vocal room accommodating errors in estimating W. So the idea is that the best separator nine is the one.

[65:00] 发言人 1: One that marks my smart, I put the largest distance between the close point and the distance buffer. And here’s my question: according to the four figures which. This is the best after this one. Which one of the four is the best?

[65:34] 发言人 1: 我不是。So the bottom right one, right? The bottom right one is the best.

[65:44] 发言人 1: So the sub second question is: is it optimal? to separate the two classes. THE BOTTOM RIGHT ONE.

[65:57] 发言人 1: Is this optimal? Or putting another way, are you able to find a better solution? Number one, please on the right side.

[66:17] 发言人 1: We can move the separating hyperplane along this direction a little bit, right? 呃，in order to increase the margin。Until there is a point supporting this market.

[66:38] 发言人 1: Right，that is，in theory，we should have margin points on both side。Otherwise, we can improve the margin by moving this hyper plan to the direction where on that side there is no margin point. right.

[67:01] 发言人 1: So the solution by symmetry, this intuition is formalized as symmetry by symmetry. There should be at least one margin point on each side of the boundary. And the point of marginal support support vectors and post support or define the margin.

[67:28] 发言人 1: So here we know. 协会。Introduce the maximum margin principle and first things first we need to compute the margin.

[67:43] 发言人 1: We need to we need to know how to compute the margin. And then we are able to maximize the margin. So what is margin? A distance between double transfer x plus b is zero and the point x five. 视频。Then this line I will transport it plus B zero zero and this line will work this direction and have this particular x.

[68:20] 发言人 1: 啊，我们 compute margin，which is distance between the point and this，have a plan，and you need compute this。啊 instance，我这个 ID。喂，SO first。所以，我我都是啊。W divide by its length its length is a.

[68:57] 发言人 1: 有你只能是代表。We just want to use this direction of this lens that’s why we open this. points in the. 是。

[69:23] 发言人 1: in the same direction. S W. 然后，能 we are able to move this point x。in the direction of negative. W over W squares.

[69:49] 发言人 1: 要从 W in a negative direction by the distance of D。Right to get its projection known as X R. 然后的公司 projection，remove x towards the active direction of.

[70:12] 发言人 1: Using excuse men’s business. And we we know that this expire is on the on the separate hyperplane. That’s then we have W transpose.

[70:29] 发言人 1: X minus D W over the from。class B is zero，然后加所有货币，来，咱们 here this is W transport X。minus d w transports w divide by delta of w plus b is equal zero. 产品。And here we have.

[71:03] 发言人 1: 呃，从 S squares 呃，从 D is equal to W transpose plus B right, remove。This to the right hand side and reverse the the right hand side and the w transport substances from squares to long squares divided by out long is equal to out long of w. And this are the transfers express p divided by。And here we consider this

point, this point with label y is it right? And if we consider points here corresponds to why is negative one and how to projection in this case to projection we move.

[72:05] 发言人 1: We move x towards this direction, right? Towards this direction by amount of d. So when W Y is next one this is a projection.

[72:26] 发言人 1: 所以这里，所以 on the two sides with opposite sign，所以 if we replace this here，and so for d we have d。we have b is equal to negative w transfer sex plus b right by long squares and combine these two. We have D is equal to Y times W transport X plus B divide by the normal W。好的结果有没有主观哎结果有没有那个主观。

[73:16] 发言人 1: And this recovers this distance. This is called geometric margin, the numerary, the numerary yi, w transpose x i plus b for the functional margin. And what’s the margin between w transpose x plus b zero and training data set d? And we take the minimum one.

[73:41] 发言人 1: Minimum of minimum one which is minimum over the data set one of the transport plus B right by the whole. Then the maximum solution is find by the maximum. MY RESPECT TO THE.

[74:00] 发言人 1: W B the inside of provinces is a marching respect for Chinese data set where we take out my hour W normal W because it’s independent of X and Y. I know that if we make scanning w to gamma w b to gamma b where gamma is square zero, the object functions are changed. Remember inside this one, if we change this to why. 伽马 W transpose X plus 伽马 B divide by the 伽马 W，然后从伽马 W this is equal to the gamma in the.

[74:54] 发言人 1: I 四十二。And this is known as this is known as skinny and beauty. And we can use freedom to set.

[75:09] 发言人 1: The function margin to be a distance one a true open number. Right, so why we said this equal to one? I said two, three, five.

[75:22] 发言人 1: It’s up to you. But for one, there are two reasons. One is the rotation, right? We always on the same techniques either for most comparison.

[75:34] 发言人 1: And you currently we have since the minimum is one or should be greater than equal to one, right? Y I X W. Y X Y W transpose equals transpose B square equals one 喂，衣服是 this。

[75:59] 发言人 1: If we set this remove this two here, this is one, this gives you months w b one over. 除了 one，right，this corresponds to minimum。W B L to long of W remember L to long of W we have a solution right we don’t want work with.

[76:24] 发言人 1: 出入。This WB will have L long squares with square it and square if you provide our limited derivative there is free from we want to cancel it that’s when we put it that’s when we put. ええ。

[76:47] 发言人 1: While we’re two in front. Remember the most transition the maximum value for sure which course we see these things. Was versus but info argument W that the recent must.

[77:07] 发言人 1: Essentially, we should write here as our commands. 玩去告诉汪汪，然后。therefore we arrive at the primal form hard margin as well.

[77:25] 发言人 1: the game on twenty third we optimize we minimize our two squares of w subject to function much with one. So the object minimize the universal margin is that maximum the geometric margin. The quality ensure all point are either on or outside the function margin.

[77:44] 发言人 1: The function margin is said to be a distance to one from the company or production. Same as not just requirement, we use the sign of the function or fine function. So why is.

[78:00] 发言人 1: Maximum 的 margin is good，the two we have two reasons。First, the two that is uncertain maximize margin allows most。Uncertainty make a room for w one keeping all the points pre classified the second data points are uncertain so markets margin allows most make a room of the points while keeping all points correctly classified so is hard much enough.

[78:32] 发言人 1: So here shows two events. If for now we ignore this noise point, noisy point, and then here is the margin, the decision boundary with this large margin. Once we add this single lowest point, a single example that will affect the deciding boundary substantially from the dash pipeline. To the black line with the marching becomes much smaller right from this angle you see how sensitive.

[79:21] 发言人 1: Second, for the most, it’s not a link for not an accessible data bus or mostly inaccessible. The primal form, how much this will give you low solution because there. There is no feasible solution that satisfy the criteria, and how do we resolve these two cases? We use the same linear classifier, we allow some training samples to violate the marching that are inside the marching array or even misclassified across the decision boundary.

[79:59] 发言人 1: 啊，穿为定义了 site variable for each example，可说 eye return zero，可说 eye equal zero，mean sample is outside of the market windows type。Because the equation means someplace inside the market area. We arrived at the private form software version as well.

[80:24] 发言人 1: So let’s first example quite subject to why the function art now can be this someone that this inside the monetary or even be classified where there is greater than one, yeah, this is which are zero. And we want to punish the inner sound. Of the violations.

[80:51] 发言人 1: With the C as a penalty parameter. 所以 you may as well let’s say personalize. The squares is up to you.

[81:06] 发言人 1: Right, you can realize squares for sure. This give you another version of arguments. But in as well, we we choose to use a linear linear linear penalty.

[81:17] 发言人 1: And smaller values means a lot more variations, and larger values means don’t allow. 嗯， violations and the effect of C with the increase of C，you see how the margin error decrease in order to penalize violations. 哎，我们不要 to the point be in the margin。And how we do that, we decrease the marching area.

[81:55] 发言人 1: And in the case of prime for massage with soft margin is equivalent to minimize the following function. So let’ s derive it. So here for this in condition, well, the eyes return to zero. For this, we also have to see I is greater than or equal to one minus.

[82:22] 发言人 1: Why I are the transpose Xi plus B right? Move this to the left hand side，move this to the right hand side。And then cosine is greater than those terms therefore cosine i should be greater than equal to the maximum of these two.

[82:57] 发言人 1: right! Is it clear?

[83:08] 发言人 1: Then we replace the CI with its no bound. And the device by sea. And this recovers this formula, one over two C built from the squares, one over two C over two squares of W, and the sum and the sum from I maximum zero one meter one meter one by I. 三。

[83:46] 发言人 1: right we recover the book and here right i see here from the side. 哎，下次少说一点。But why are we able to.

[83:59] 发言人 1: Why are we able to replace the with slow down and minimize slow down? So pictorially something something at this right so here. So my left hand represents the eye and my right hand represents the lower bound. And the first minimize the lower bound, then set because the eye two is lower bound. That’ s why is good or is it true.

[84:43] 发言人 1: And here this is just some custom right we can replace one or two thing we can observe this two in the same and write this as one overseas as you want this gives you the same say you know this relation. And here we can define why I this is F X I we can define why I can’ t say exactly ice as the eye and recover what’ s known as hinge loss where maximum returns whichever value maybe is larger. And it takes tens of years from hinge nose with them.

[85:25] 发言人 1: And here is the figure of the hinge nose. we have large linear loss. For this class, for samples, we have long inner loss.

[85:38] 发言人 1: For samples, steer the margin. We have exact zero loss on the correct side of the margin. And zero one loss, suppose logistic loss and hinge loss are complex surrogate loss of the zero one loss.

[85:53] 发言人 1: So here is the figure for zero runners. So if it’ s in. Correctly classified，we have the penalty of one，if it’ s correctly classified，have penalty of zero，right?

[86:07] 发言人 1: This is C，this is L zero one C，the average zero one loss over get the easy value classification error rate。So this is a most function with the like to minimize bodies computationally visible because it has written almost. Gradient of zero almost everywhere at this point where the gradient is not defined.

[86:33] 发言人 1: Therefore, we are calling for several bit functions. And here, in this figure, we compare hinge logistic exponential and zero one loss. We add an extra exponential loss for comparison.

[86:48] 发言人 1: Because it used in the direction or other machinery are such as and boosts. As you see, the zero one was. Zero one loss，the expansion loss，the hinge loss all go across this point zero one。啊 remember the logistics loss.

[87:08] 发言人 1: It’s not one over. 你 to the negative C right. 干啥收衣服。I draw logistic loss in a blank curve like this.

[87:23] 发言人 1: Can you infer the base of the log? what’s the base of the dome in this case? If the base log is two, this majestic loss will also go across this point zero one, and all those functions are anchored at point zero one. And in this control, they know using natural log, which result in zero point six.

[88:09] 发言人 1: 单车 multi. Class as well, so one versus one multi-class classification. We train binary classifiers on or pairs of classes.

[88:28] 发言人 1: So for three classes, example, have one versus two, one versus three, two versus three to label a sample. We pick the class with most votes among. Or battery passengers the problem numbers one is very slow when. the number of classes is large and if there are k classes we need to train k minus one battery we need to train c k two. 可以还是可以卖出。Over two very class virus, one versus best math class classification between our space very class virus one versus two three one as positive two as negative and so on to label sample pick the class with largest margin here what’s this margin? 呃，有 functional market。

[89:39] 发言人 1: 循环的意思，有没有循环区，我发布循环区。

[90:05] 发言人 1: So what’s the functional margin here? So this is the functional margin.

[90:18] 发言人 1: So if we want to use your match machine, how are going to modify it? Divide by what? Divide by the L two norm of w, right? So which one you think is more reasonable?

[90:56] 发言人 1: So here, so I, I, I can give you one personal story. So when I prepare the size, I think five or six zero. There’s no, there’s no, there’s no chat p p t.

[91:10] 发言人 1: Right? I don’t know why. I, I don’t, I have no intuition which we should use function marking or duration marking. But we we know that at that time there are a lot of virtual machinery packages right so i want to know how those types in particular this decision was then i i spent this half an hour to locate the specific posting how they implement this difference.

[91:42] 发言人 1: 嗯，还有第三，第三，但是现在，with A I A I tools，you may locate the go sleep within this ten seconds。And read read read the code and even ask the. 嗯，嗯，嗯，嗯，嗯，models，which one you think is more reasonable，we use function much，or we use geometry much。果然。

[92:22] 发言人 1: And there are the decision boundary for each battery classifier. So this is, let’s say, w zero. This is for w one. This is w two.

[92:40] 发言人 1: And we combine those three according to the decision decision I came in the previous nights in order to give this his wife media is in the conference.

Alright，so now let’s do summary as a class bar，it will have well。 dealing with a linear function w transpose x plus b, and this is the same linear function used in.

[93:19] 发言人 1: Same linear function used in the objective refresh and given a new sample we predicted as one star is equal to sign of the transport study remember. Definition of sine function, right? So this is x, this is sine of x, we have.

[93:43] 发言人 1: Positive gives you one, negative gives you negative one. And we have zero on specify. And training is to maximize the margin of the training data, and maximize the. between points and the decision boundary. And in in the case of primal form hard margin we don’t allow any valuations and if we apply this to the most.

[94:17] 发言人 1: to the leader not circle data we cannot get a single face resolution that’s why we move to soft margin case we allow some training samples to violate the margin or even go across the decision boundary where the corresponding side rapper say i. is greater than one greater than one means one matter success this are zero this are zero means misclassified we use cross validation as we introduced in the previous nature to pick the hyperparameter see. 啊。

[95:01] 发言人 1: Linear cluster，we separate the data using a linear surface hyperplane，y is equal to sine w translate x plus b。We have two formulations，why is not just refreshing，not just integration is a. probability approach。Because he directed models the posterior probability or I P of why. 嗯，对，本来。using what function.

[95:48] 发言人 1: Using the same voice function right? so what’s the sigmoid function. As one over one plus into negative z，a signal function。It marks as a poor bit of the data, and what’s the loss function associate with logistic regression?

[96:22] 发言人 1: 那时候 just goes right。Let’s knock one plus one to the next day.

[96:39] 发言人 1: And we have already show that. So for those comparison, we would better use this two because all those functions anchored at one zero one. Now this also justifies why we want to the functional margin.

[96:58] 发言人 1: of the hinges. to be one. So remember that, so this is a hinge loss.

[97:10] 发言人 1: So this is one, this is one, so this is the hinge nose, right? If we set let’s say the functional margin. All function must be greater equal to let’s say two.

[97:27] 发言人 1: You can’t do that, but now the most function moves to the right. And they are vote, it’s for anchor like zero one, it’s anchor like zero two. Alright, and then that’s why，then you need to，it’s better to set the minimum functional margin to be one。

[97:56] 发言人 1: So for support vector machine, it is surely a. geometric approach. As maximize the margin of the hyperplane, how to compute the margin, the geometric margin, right? The geometric margins, why I? So here we use the notation system of zero one. So in the next sequence, we will get you through the zero one combination.

[98:33] 发言人 1: W transpose x i plus b divided by the global public that’s your match margin the numerator is about margin of hydrocarbon and we do have

what’s known as hinge loss socialism. Integrals at maximum zero one minus three which prove you should here if we use a functional model distance two this one becomes two. 他是供给款是。

[99:13] 发言人 1: For loss functions adjust loss push the classification angry as far as possible for all points. Why for all points remember logistic loss will give an all points some positive penalty. Even points are away from the distance because this is always gonna be greater than zero.

[99:51] 发言人 1: That’s why push the classification boundary as far as possible for all points. All right, support of that machine on the other hand ensures a margin. Or one between boundary and close point. It can give zero penalty to points that are far away from decision boundary.

[100:23] 发言人 1: 六。

[100:30] 发言人 1: So advantage is s one the s one works well on high dimensional features when n is large. So if the freight is of high dimensions, more or less is likely to become the data is likely to become almost linear separable or linear separable. That’s why it has in previous state has a no generalization here.

[101:00] 发言人 1: 咳咳，或者看基因化最近 prevail，咳咳，我们 just refreshing on the other hand。Has well defined probabilities that is p or y even x device through logistic function. And this advantage is the decision surface can only be linear.

[101:25] 发言人 1: Next actually we will see how to deal with nonlinear decision surface through. The introduction of the kernel formulation over the kernel trick, and they will introduce through the dual formulation with with as well. beautiful.

[101:47] 发言人 1: And in this case, in this literature, we only introduce the platform. Alright，any question on the supercomputer machine，the problem for power battery and the problem for software market。Alright, lecture three point three.

[102:11] 发言人 1: The TA team has prepare you guys a very bit. I buy some notebook for you guys to implement play with and understand logistics regression. And in this case.

[102:24] 发言人 1: We work with the location system, see here. 都推进 system。Why is there one? No, this is not next to one one.

[102:38] 发言人 1: And here we can arrive at what’s known as the central piano. Yeah. This is cross entropy loss. And you can you take derivative with the henchmen and cross entropy they will arrive at the same vector.

[103:01] 发言人 1: And I will put, I will update the lecture loads to work with this, this two formulations and compare the present and you see the equivalents of two formulations. And here’s a neat implementation. And this is brilliant.

[103:24] 发言人 1: And I also introduce some evaluation criteria precision record and accuracy and if one you have good understanding of how those metrics are defined implemented. And different metrics and processes, that different aspects. Alright, that’s all for today’s security questions.

[104:03] 发言人 1: All right，so two things are cost logistics，so we we we do have a to world。After class right i begin at eleven thirty so one of my postdocs doctor wang zhen will lead the tutorials and so i was notified that we will have a makeup lecture on this sunday right. September twentieth.

[104:36] 发言人 1: Because we have a long middle festival national holidays. Right, so I will see you. This something at the same time, alright? That’s all.

[105:03] 发言人 1: 我知道周周日不补课吗周日啊嗯对对但是他说。September thirty. 你听到的是什么 September 吗？没有啊，因为我们是 TWENTY 吧，二十五号是中秋，所以我们呃得二十号在哦 SEPTEMBER THIRTEEN THIRTY，三十号，三十号有一个，三十号有一个是吧？没有啊。哎呀。